Sanjay Jacob

sanjayjacob19@gmail.com ● **+353 87 034 1516** ● SSJJCC-oss · GitHub ● www.linkedin.com/in/sanjay-jacob-3b66b01b1 ● Eligible to work full-time in Ireland (Stamp 1G)

Profile

MEng Graduate in Computer Vision and Artificial Intelligence from the University of Limerick, with a First-Class BSc in Computer Science and an MSc in Artificial Intelligence from the University of Kent. Combines strong academic foundations with industry experience as a JavaScript Developer at CaseWare UK, where I built production-level modules for a global financial SaaS platform and AI Intern at Sureicon IT Solutions LLP. Technical expertise spans deep learning, NLP, and full-stack development, with hands-on experience deploying AI models on edge hardware. Dissertation focused on privacy-preserving edge AI for health analytics.
I'm open to graduate and entry level roles across AI Engineering, Machine Learning, Computer Vision and Data Science.

Work experience

Sureicon IT Solutions LLP: Intern AI Developer **September 2024 to August 2025**
Sureicon focuses on digital transformation and business software solutions by integrating AI automation, serving sectors such as general trading, manufacturing, distribution, and oil & gas.
Website: https://www.sureicon.com/

Integrated natural language processing (NLP) chatbots into web and client applications to automate customer interactions, support ticketing, and lead qualification. Automating manual data entry, processing invoices, and structuring unstructured data within enterprise applications.

CaseWare UK Ltd: JavaScript Developer **June 2021 to May 2022**
CaseWare is a global provider of cloud-enabled audit, financial reporting, and data analytics software used by accounting firms and enterprises.
Website: https://www.caseware.com/

Developed and deployed JavaScript modules for a financial SaaS platform serving enterprise clients globally. Collaborated in Agile sprints contributing to sprint planning, code reviews, and production debugging. Improved platform performance and reliability across production systems used by 400,000+ accounting professionals worldwide. Built scalable web application features in a professional team environment.

Education

MEng Computer Vision & Artificial Intelligence (GPA 2:1)
University of Limerick, Ireland (**September 2025 to August 2026**) Completed modules in machine learning, deep learning for visual data, and geometric and machine vision, with a focus on deploying deep learning models on edge devices. Studied the ethics and governance of AI systems. Dissertation: a privacy-preserving edge AI monitoring system running on Raspberry Pi (see Projects). Took part in hackathons, building working prototypes under time pressure.

MSc Artificial Intelligence (Merit GPA 2:1)
University of Kent, Canterbury, UK **(September 2023 to August 2024)**: Built a strong foundation in artificial intelligence, covering both the theory behind how AI systems work and the practical skills to build them. Gained hands-on experience developing and training machine learning and deep learning models using PyTorch and TensorFlow, working across areas such as computer system, NLP, and data analytics.

BSc Honours Computer Science with a Year in Industry (First Class Honours)
University of Kent, Canterbury, UK (**September 2019 to May 2023**): Built a strong foundation in computer science covering distributed systems, networking, blockchain, IoT, and data-driven problem solving. Gained real-world software development experience during a year in industry at CaseWare UK.

Projects

- **Portfolio:** https://sanjayjacob.vercel.app/
- **Elder Monitoring System:** An End-to-End Full-Stack Framework for Privacy-Preserving Edge Monitoring and Predictive Health Analysis. Built using Python, MediaPipe, YOLOv8, Flask, Raspberry Pi, and edge AI deployment. Live dashboard https://iseehome.ai
- **WhatsApp AI Ordering Agent**: Built a Google ADK and Gemini agent that takes restaurant orders over WhatsApp. It checks live stock, records orders and runs scheduled special-offer campaigns to opted-in customers: https://github.com/Appootty/restaurant-order-agent
- **Language-Facilitated Robot Task Planning** (**MSc Final Project**) - Webots simulation with OpenAI-powered voice control + A* path planning. https://github.com/SSJJCC-oss/RobotWheelChair
- **Credit & Collections Copilot (In Progress):** Multi-agent Google ADK + Gemini copilot that helps wholesale distributors chase overdue payments and approve credit orders, with explainable rule-based decisions, human-approved reminders and safe read-only data Q&A. https://github.com/SSJJCC-oss/credit-collection-copilot

Technical skills

Skills:

- **Programming:** Python, Java, C++, C#, JavaScript (HTML, CSS, Node.js, PHP basics)
- **Machine Learning / AI:** Large Language Models, Transformer Models, Retrieval Augmented Generation, Natural Language Processing, MediaPipe, YOLOv8, TensorFlow, OpenCV, PyTorch
- **Agentic AI:** Google ADK, Gemini, LLM tool/function calling, prompt design
- **Data Science:** Pandas, NumPy, Matplotlib
- **Computer Vision:** YOLOv8, real-time video processing, edge AI deployment, Raspberry Pi, embedded systems
- **Cloud & Tools:** AWS (SageMaker basics), Google Cloud AI, Git/GitHub, Docker, FastAPI, Google Cloud Run, Twilio

Interests

- **Speed Cubing:** 14+ years of experience, competed in 20 competitions across 3 countries, winning 4 medals.
- **Driving & Travel:** Passionate about driving and exploring new places, with experience driving across diverse environments and conditions, reflecting adaptability and confidence in unfamiliar situations